MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP(HONOURS)REGULAREXAMINATIONOCTOBER 2026FIFTH SEMESTER

DisciplineSpecificCourse(DSC)-MG5DSCFSQ300-TECHNOLOGYOFMEAT,FISH,EGGANDPOULTRY

(2024ADMISSIONONWARDS)

Duration:1 Hour 30 Minutes MaximumMarks:50

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Skill(S),Interest(I) andAppreciation(Ap)

Studentsshouldattemptatleastonequestionfromeachcourseoutcometoenhancetheiroverall outcome attainability.

Part A

MCQ

Answerallquestions

1. C Evisceration
2. A $121^{O}C$
3. A Repeated freezing and thawing
4. A Sarcomere
5. A Remove internal organs
6. A Fin fish and shellfish
7. B Salami
8. C Mild fresh odour
9. B Shell
10. D Enzyme lose activity and metabolism stops

Part B – Short Answers (5 marks each)

11. Summarize the changes in meat colour and flavour during the post-mortem period.

A. Changes in meat colour

1. Immediately after slaughter, meat may appear purplish-red due to deoxymyoglobin.
2. When exposed to oxygen, myoglobin forms oxymyoglobin, producing a bright cherry-red colour.
3. During storage, oxymyoglobin is oxidized to metmyoglobin, which gives meat a brown colour.
4. The rate of colour change depends on oxygen availability, pH, temperature and storage conditions.
5. Microbial growth and prolonged storage can cause discoloration.

B. Changes in meat flavour

1. After slaughter, muscle glycogen is converted into lactic acid.
2. ATP breakdown and biochemical changes contribute to the development of meat flavour.
3. During ageing, enzymes break down proteins and other muscle components, producing flavour precursors.
4. Cooking produces characteristic meat flavour through Maillard reactions and fat oxidation.
5. Spoilage microorganisms may produce unpleasant odours and flavours.

12. Explain the relationship between albumen quality and egg freshness.

1. Fresh eggs have thick, firm albumen (egg white) with a high proportion of thick albumen.
2. During storage, carbon dioxide escapes through the shell, causing the albumen pH to increase.
3. The albumen becomes thinner and more watery as its protein structure changes.
4. The thick albumen height decreases, resulting in a lower Haugh unit value.
5. Therefore, higher Haugh unit values generally indicate better albumen quality and greater freshness.

Haugh unit: An index used to evaluate egg interior quality based mainly on albumen height and egg weight.

13. Describe the advantages and disadvantages of ohmic heating.

Advantages

1. Rapid and uniform heating of suitable food materials.
2. Shorter processing time and improved energy efficiency under suitable conditions.
3. Better retention of colour, flavour, texture and nutrients in some foods.
4. Reduced damage caused by prolonged heating.
5. Suitable for foods containing particles suspended in liquids, such as soups and sauces.

Disadvantages

1. High initial investment and equipment cost.
2. Electrical conductivity varies between food components, which can cause uneven heating.
3. Electrode corrosion and electrochemical reactions may occur.
4. It requires careful control of voltage, current and processing conditions.
5. It is less suitable for foods with very low or highly variable electrical conductivity.

14. Identify the procedures involved in ante-mortem examination before slaughter**.**

1. Document verification: Check animal identification, ownership and relevant movement or health records.
2. Observation at rest: Examine the animal for abnormal posture, behaviour, breathing and signs of illness.

3. Examination in motion: Observe walking, gait and signs of lameness.
4. Clinical examination: Check visible body condition, eyes, skin, mucous membranes and abnormal discharges.
5. Identification of abnormalities: Look for fever, injuries, swelling, weakness or suspected infectious disease.
6. Segregation: Separate sick, injured or suspect animals for further examination.
7. Decision: Classify animals as fit for slaughter, requiring further examination, or unfit for slaughter according to applicable regulations.
8. Humane handling: Ensure proper rest, water and welfare during lairage.

15. Classify meat based on source and characteristics with suitable examples.

A. Based on source

1. Beef: Meat obtained from cattle.
2. Veal: Meat obtained from young calves.
3. Pork: Meat obtained from pigs.
4. Mutton: Meat obtained from adult sheep.
5. Lamb: Meat obtained from young sheep.
6. Chevon: Meat obtained from goats.
7. Poultry meat: Meat obtained from chicken, turkey and ducks.
8. Game meat: Meat obtained from wild animals, such as deer.

B. Based on colour and characteristics

1. Red meat: Rich in myoglobin; examples include beef, mutton and chevon.
2. White meat: Generally contains less myoglobin; examples include chicken breast and turkey breast.
3. Processed meat: Meat preserved by curing, smoking, fermentation or other processing; examples include ham, bacon and sausages.

16. Define vacuum packaging and explain its principle.

Definition: Vacuum packaging is a food preservation method in which air is removed from the package before it is hermetically sealed.

Principle

1. Meat is placed inside a suitable, low-permeability packaging material.
2. Air is removed using a vacuum packaging machine.
3. The package is sealed to prevent air from entering.
4. Reduced oxygen slows the growth of many aerobic spoilage microorganisms.
5. It reduces oxidative rancidity, surface discoloration and moisture loss.
6. It extends the shelf life of meat when combined with proper refrigeration.

Limitation: Vacuum packaging does not sterilize meat. Refrigeration and appropriate food-safety controls are essential because some anaerobic microorganisms can grow in low-oxygen conditions.

Part C – Essay Answers (10 marks each)

17. Discuss the spray-drying process used for producing egg powder.

Definition: Spray drying is a process in which liquid egg is converted into powder by spraying it into a stream of hot air.

Steps involved

1. Selection and cleaning: Fresh eggs of good quality are selected and cleaned appropriately.
2. Breaking: Eggs are mechanically broken, and the contents are collected.
3. Separation: The egg contents may be separated into yolk, albumen or whole egg according to the required product.
4. Filtration: Shell fragments and other foreign materials are removed.
5. Pasteurization: Liquid egg is pasteurized using a validated time-temperature treatment to reduce pathogenic microorganisms.
6. Concentration: Some water may be removed before drying to improve process efficiency.
7. Atomization: Liquid egg is sprayed through a nozzle or rotary atomizer into fine droplets.
8. Drying: Hot air rapidly evaporates moisture from the droplets inside the drying chamber.
9. Powder collection: The dried particles are separated from the exhaust air using cyclones or suitable collection systems.
10. Cooling and packaging: The powder is cooled and packed in moisture-resistant, airtight packaging.

Flow chart

Egg selection → Breaking → Filtration → Pasteurization → Concentration (optional) → Atomization → Spray drying → Powder collection → Cooling → Packaging.

Advantages

- Long shelf life when properly processed and stored.
- Reduced weight and volume.
- Convenient transportation and storage.
- Easy to measure, handle and reconstitute.
- Useful in bakeries, confectionery, pasta and food manufacturing.

Limitations

- High initial equipment cost.
- Heat may cause protein denaturation or functional changes.

- Moisture absorption can cause caking during storage.
- Improper processing may reduce product quality or safety.

18. Discuss the major meat by-products obtained during animal slaughtering.

Meat by-products are the useful materials obtained from animals during slaughtering, apart from the main carcass meat.

1. Edible by-products

- Liver: Rich in iron, vitamins and protein.
- Heart: Used as food and contains high-quality protein.
- Kidneys: Used for human consumption after proper inspection.
- Tongue and brain: Consumed in some regions.
- Tripe: The edible stomach lining of cattle and other ruminants.
- Blood: Used in certain food products where permitted.

2. Hides and skins

- Processed into leather.
- Used to manufacture shoes, bags, belts and other leather products.

3. Bones

- Used to produce bone meal, gelatin and certain mineral products.
- May be processed into animal feed ingredients where legally permitted.

4. Fat and adipose tissue

- Rendered to obtain tallow or lard.
- Used in soap, candles, industrial products and certain food applications.

5. Intestines

- Processed into natural sausage casings.
- Some intestinal tissues are used in the production of pharmaceutical materials.

6. Blood

- Can be processed into blood meal for animal feed where permitted.
- Plasma and other fractions may have food or industrial applications.

7. Glands and organs

- Certain glands and tissues are processed to obtain enzymes, hormones and pharmaceutical ingredients.

8. Hair, horns and hooves

- May be used in fertilizers, protein meals and selected industrial products.

9. Waste materials

- Inedible tissues and other suitable organic residues can be processed through rendering or other approved treatment methods.
- Materials classified as specified risk material or condemned waste must be handled and disposed of according to applicable regulations.

Importance of by-product utilization

- Reduces slaughterhouse waste and environmental pollution.
- Provides additional income to the meat industry.
- Recovers useful nutrients and industrial raw materials.
- Supports more efficient and sustainable use of animal resources.

19. Summarize the preparation, nutritional significance and application of fish protein concentrate (FPC).

Definition: Fish protein concentrate (FPC) is a concentrated protein product obtained from edible fish or suitable fish material by removing much of the water, fat and other non-protein components.

A. Preparation

1. Raw material selection: Select fresh fish or suitable fish material.
2. Washing and cleaning: Remove dirt, scales and unwanted material.
3. Size reduction: Mince the fish to increase processing efficiency.
4. Extraction: Use an appropriate food-grade process, often involving solvents such as ethanol or isopropanol, to remove fat and odorous substances.
5. Separation: Separate the protein-rich material from the extraction liquid.
6. Solvent removal: Remove residual solvent using suitable controlled processing.
7. Drying: Dry the material to obtain a low-moisture product.
8. Grinding and sieving: Convert the dried material into a uniform powder.
9. Packaging: Pack in airtight, moisture-resistant packaging and store under suitable conditions.

Flow chart

Fresh fish → Washing → Cleaning → Mince preparation → Fat and odour removal → Separation → Drying → Grinding → Packaging.

B. Nutritional significance

1. Contains a high concentration of protein.
2. Supplies essential amino acids required for growth and tissue repair.
3. Provides protein in a compact, low-moisture form.
4. May retain minerals and other nutrients, depending on processing.
5. Can help improve the protein content of foods.

C. Applications

1. Used as a protein supplement in foods.
2. Added to soups, sauces and savoury products.
3. Used in fortified foods and nutritional formulations.
4. Applied in bakery and cereal-based products.
5. Used in selected animal feed formulations where appropriate.

D. Limitations

- May develop a fishy flavour or odour if processing is inadequate.
- Excessive heat may reduce protein functionality.
- Some people may experience allergic reactions to fish proteins.
- Production cost and solvent recovery requirements may be high.

20. Explain the relationship between oxygen availability, microbial growth, lipid oxidation and meat colour in vacuum-packaged meat.

Definition: Vacuum packaging removes most of the air surrounding meat and seals it in a package with low oxygen permeability.

1. Effect of oxygen availability on meat colour

- Myoglobin is the principal pigment responsible for meat colour.
- In the absence of oxygen, deoxymyoglobin gives meat a purplish-red appearance.
- When the package is opened and meat is exposed to oxygen, oxymyoglobin forms and produces a brighter red colour.
- During prolonged storage, metmyoglobin may accumulate and produce brown discoloration.
- Therefore, purple meat in an unopened vacuum pack is not necessarily spoiled.

2. Effect on microbial growth

- Removing oxygen inhibits many aerobic spoilage bacteria, including Pseudomonas species.
- It may favour the relative growth of lactic acid bacteria and other organisms that tolerate low oxygen.
- Vacuum packaging does not destroy microorganisms already present in the meat.
- Certain anaerobic pathogens, including Clostridium botulinum, may present a risk under unsuitable storage conditions.

- Proper refrigeration, hygiene, shelf-life control and validated processing are essential.

3. Effect on lipid oxidation

- Oxygen promotes the oxidation of unsaturated fatty acids in meat.
- Lipid oxidation produces peroxides and secondary compounds that can cause rancid odours and flavours.
- Vacuum packaging reduces oxygen exposure and generally slows lipid oxidation.
- Consequently, it can help preserve flavour and nutritional quality during storage.
- Oxidation is reduced, not completely prevented.

4. Relationship between oxygen and colour stability

- Oxygen exposure affects the chemical state of myoglobin.
- High oxygen exposure may initially improve the bright red appearance but can also promote oxidative deterioration.
- Low oxygen maintains deoxymyoglobin and limits oxidation, producing a darker purple-red colour.
- Colour changes must be assessed along with odour, texture, package condition and microbiological safety.

5. Overall advantages of vacuum packaging

- Extends shelf life under suitable refrigeration.
- Reduces oxidative rancidity.
- Minimizes moisture loss and contamination after packaging.
- Preserves flavour and quality for a longer period.
- Facilitates storage and transportation.